REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Methods of Science Practice Questions

Grade 9 Advanced | Module 1 Lesson 1 | Methods of Science

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Compare evidence, repeat trials and change only one independent variable.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 3–8.

P01. Which statement best describes a scientific hypothesis?

A. An unchangeable belief B. A testable proposed explanation
C. A measurement unit D. A final proof

P02. Why do scientists repeat experiments?

A. To guarantee a preferred answer B. To avoid collecting data
C. To check whether results are reproducible D. To remove the need for controls

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P03. What is the main purpose of a scientific model?

A. To reproduce every detail exactly B. To replace all observations
C. To prove that testing is unnecessary D. To represent and explain selected features of a system

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P04. A scientific theory is best described as:

A. An explanation supported by a broad body of evidence B. An unsupported guess
C. A single experimental reading D. A mathematical unit

P05. What should a scientist do when reliable evidence contradicts a model?

A. Hide the evidence B. Reconsider or revise the model
C. Change the measurements without justification D. Stop all investigation

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P06. A class tests 4 identical ramps and repeats each test 5 times. How many measurements are collected?

A. 40 measurements B. 10 measurements
C. 20 measurements D. -20 measurements

P07. Three repeated measurements are 12, 14 and 16 s. Calculate their mean.

A. 28 s B. 7 s
C. -14 s D. 14 s

P08. A control test takes 26 s and a test after changing one factor takes 10 s. By how many seconds did the measured time decrease?

A. 16 s B. 32 s
C. 8 s D. -16 s

P09. An investigation compares 4 surface materials using 5 repeat trials on each. How many trials are needed in total?

A. 40 trials B. 20 trials
C. 10 trials D. -20 trials

P10. Repeated readings are 22, 24 and 26 cm. Calculate the range of the readings.

A. 8 cm B. 2 cm
C. 4 cm D. -4 cm

P11. State the main subject matter of physics.

P12. Distinguish an observation from an inference.

P13. Distinguish a scientific law from a scientific theory.

P14. Why is peer review useful?

P15. What makes a hypothesis scientifically useful?

P16. A class tests 4 identical ramps and repeats each test 6 times. How many measurements are collected?

P17. A class tests 4 identical ramps and repeats each test 7 times. How many measurements are collected?

P18. Three repeated measurements are 13, 15 and 17 s. Calculate their mean.

P19. Three repeated measurements are 14, 16 and 18 s. Calculate their mean.

P20. A control test takes 29 s and a test after changing one factor takes 11 s. By how many seconds did the measured time decrease?

P21. A control test takes 32 s and a test after changing one factor takes 12 s. By how many seconds did the measured time decrease?

P22. An investigation compares 5 surface materials using 5 repeat trials on each. How many trials are needed in total?

P23. An investigation compares 6 surface materials using 5 repeat trials on each. How many trials are needed in total?

P24. Repeated readings are 23, 25 and 27 cm. Calculate the range of the readings.

P25. Repeated readings are 24, 26 and 28 cm. Calculate the range of the readings.

P26. A student tests whether ramp height changes travel time, but changes both height and surface material. Evaluate the method.

P27. A student says one result proves a hypothesis forever. Evaluate the claim.

P28. Design a fair test of how surface roughness affects the distance a toy car travels.

P29. Two groups obtain different results using the same method. Suggest two constructive next steps.

P30. A model predicts the correct trend but not the exact measured values. Must it be discarded immediately? Explain.